

The Tangent Bundle and Partitions of Unity

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The Tangent Bundle

Let M be a smooth manifold.

The **tangent bundle** of M is the union of all the tangent spaces of M :

$$TM := \bigcup_{p \in M} T_p M$$

Here the union is understood as a disjoint union. That is, tangent vectors based at different points are regarded as distinct, even if they have the same coordinate expression.



The Tangent Bundle

If $(U, \phi) = (U, x^1, \dots, x^n)$ is a coordinate chart on M , let

$$TU = \bigcup_{p \in U} T_p U = \bigcup_{p \in U} T_p M$$

There is a map $\pi : TM \rightarrow M$ given by $\pi(v) = p$ if $v \in T_p M$.

There is a bijection

$$\begin{aligned}\tilde{\phi} : TU &\rightarrow \phi(U) \times \mathbb{R}^n \subset \mathbb{R}^{2n} \\ v &\mapsto (\bar{x}^1, \dots, \bar{x}^n, c^1, \dots, c^n)(v)\end{aligned}$$

where $\bar{x}^i = x^i \circ \pi$, $v = \sum_i c^i(v) \partial / \partial x^i|_p$, with inverse

$$(\phi(p), c^1, \dots, c^n) \mapsto \sum c^i \frac{\partial}{\partial x^i} \Big|_p$$

The Topology of the Tangent Bundle

Since $\tilde{\phi} : TU \rightarrow \phi(U) \times \mathbb{R}^n \subset \mathbb{R}^{2n}$ is a bijection, we can transfer the topology of $\phi(U) \times \mathbb{R}^n$ to TU .

- A set A in TU is open if and only if $\tilde{\phi}(A)$ is open in $\phi(U) \times \mathbb{R}^n$.
- TU with the topology induced by $\tilde{\phi}$, is homeomorphic to $\phi(U) \times \mathbb{R}^n$.

The Topology of the Tangent Bundle

Let $\mathcal{B}$ be the collection of all open subsets of $T(U_\alpha)$ as U_α runs over all coordinate open sets in M :

$$\mathcal{B} = \bigcup_{\alpha} \{ A \mid A \text{ open in } T(U_\alpha), U_\alpha \text{ a coordinate open set in } M \}.$$

We can check that $\mathcal{B}$ can be a basis for some topology on TM .

- $TM = \bigcup_{A \in \mathcal{B}} A$
- Let U and V be coordinate open sets in a manifold M . If A is open in TU and B is open in TV , then $A \cap B$ is open in $T(U \cap V)$.

The Topology of the Tangent Bundle

Lemma 12.2 A manifold M has a countable basis consisting of coordinate open sets.

Proof.

Let $\{(U_\alpha, \phi_\alpha)\}$ be the maximal atlas on M and $\mathcal{B} = \{B_i\}$ a countable basis for M . For each coordinate open set U_α and point $p \in U_\alpha$, choose a basic open set $B_{p,\alpha} \in \mathcal{B}$ such that

$$p \in B_{p,\alpha} \subset U_\alpha.$$

For any open set U in M and a point $p \in U$, there is a coordinate open set U_α such that

$$p \in U_\alpha \subset U.$$

Hence,

$$p \in B_{p,\alpha} \subset U,$$

which shows that $\{B_{p,\alpha}\}$ is a countable basis for M . □

The Topology of the Tangent Bundle

Proposition 12.3

The tangent bundle TM of a manifold M is second countable.

Proof.

Let $\{U_i\}_{i=1}^{\infty}$ be a countable basis for M consisting of coordinate open sets. Let ϕ_i be the coordinate map on U_i . Since TU_i is homeomorphic to the open subset $\phi_i(U_i) \times \mathbb{R}^n$ of $\mathbb{R}^{2n}$ and since any subset of a Euclidean space is second countable, TU_i is second countable. For each i , choose a countable basis $\{B_{i,j}\}_{j=1}^{\infty}$ for TU_i . Then $\{B_{i,j}\}_{i,j=1}^{\infty}$ is a countable basis for the tangent bundle TM . □

The Topology of the Tangent Bundle

Proposition 12.4

The tangent bundle TM of a manifold M is Hausdorff.

Proof.

Let $v, w \in TM$ with $v \neq w$. Suppose $v \in T_pM$ and $w \in T_qM$.

If $p \neq q$, there exist disjoint open sets $U, V \subset M$ such that $p \in U$ and $q \in V$. Then TU and TV are open subsets of TM , with $v \in TU$ and $w \in TV$. Moreover,

$$TU \cap TV = T(U \cap V) = \emptyset,$$

so v and w have disjoint neighborhoods in TM .

Now suppose $p = q$. Choose a coordinate open set U containing p . Then $v, w \in TU$, and since TU is homeomorphic to an open subset of $\mathbb{R}^{2n}$, it is Hausdorff. Hence there exist disjoint open sets $A, B \subset TU$ such that $v \in A$ and $w \in B$. Since A and B are open in TU , they are open in TM . Thus v and w have disjoint neighborhoods in TM .



The Manifold Structure on the Tangent Bundle

Claim

If $\{(U_\alpha, \phi_\alpha)\}$ is a C^∞ atlas for M , then $\{(TU_\alpha, \tilde{\phi}_\alpha)\}$ is a C^∞ atlas for the tangent bundle TM .

- $TM = \bigcup_\alpha TU_\alpha$
- It remains to check that on $(TU_\alpha) \cap (TU_\beta)$, $\tilde{\phi}_\alpha$ and $\tilde{\phi}_\beta$ are C^∞ compatible.

The Manifold Structure on the Tangent Bundle

Let two charts $(U_\alpha, \phi_\alpha) = (U_\alpha, x^1, \dots, x^n)$ and $(U_\beta, \phi_\beta) = (U_\beta, y^1, \dots, y^n)$.

Then the transition map

$$\tilde{\phi}_\beta \circ \tilde{\phi}_\alpha^{-1} : \phi_\alpha(U_\alpha \cap U_\beta) \times \mathbb{R}^n \longrightarrow \phi_\beta(U_\alpha \cap U_\beta) \times \mathbb{R}^n$$

is given by

$$(\phi_\alpha(p), a^1, \dots, a^n) \longmapsto \left(p, \sum_j a^j \frac{\partial}{\partial x^j} \Big|_p \right) \longmapsto ((\phi_\beta \circ \phi_\alpha^{-1})(\phi_\alpha(p)), b^1, \dots, b^n),$$

where $\sum a^j \frac{\partial}{\partial x^j} \Big|_p = \sum b^i \frac{\partial}{\partial y^i} \Big|_p$,

$$b^i = \sum_j a^j \frac{\partial y^i}{\partial x^j}(p) = \sum_j a^j \frac{\partial (\phi_\beta \circ \phi_\alpha^{-1})^i}{\partial x^j}(\phi_\alpha(p)).$$

Therefore $\tilde{\phi}_\beta \circ \tilde{\phi}_\alpha^{-1}$ is C^∞ .

Vector Bundles

Given any map $\pi : E \rightarrow M$, we call the inverse image

$$\pi^{-1}(p) := \pi^{-1}(\{p\})$$

of a point $p \in M$ the **fiber** at p . The fiber at p is often written E_p .

For any two maps $\pi : E \rightarrow M$ and $\pi' : E' \rightarrow M$ with the same target space M , a map $\phi : E \rightarrow E'$ is said to be **fiber-preserving** if

$$\phi(E_p) \subset E'_p$$

for all $p \in M$.

Vector Bundles

A C^∞ **vector bundle** of rank r is a triple (E, M, π) consisting of manifolds E and M and a surjective smooth map $\pi : E \rightarrow M$ that is locally trivial of rank r .

π is locally trivial of rank r if

- each fiber $\pi^{-1}(p)$ has the structure of a vector space of dimension r
- for each $p \in M$, there are an open neighborhood U of p and a fiber-preserving diffeomorphism

$$\phi : \pi^{-1}(U) \longrightarrow U \times \mathbb{R}^r$$

such that for every $q \in U$ the restriction

$$\phi|_{\pi^{-1}(q)} : \pi^{-1}(q) \longrightarrow \{q\} \times \mathbb{R}^r$$

is a linear isomorphism. Such an open set U is called a *trivializing open set* for E , and ϕ is called a *trivialization* of E over U .

Vector Bundles

Example 1 For the tangent bundle TM of a smooth manifold M , (TM, M, π) is a vector bundle where $\pi(v) = p, v \in T_pM$.

- $\pi^{-1}(p) = T_pM$
- for each $p \in M$, there is a chart (U, ψ) at p

$$\phi := (\psi^{-1} \times \mathbf{1}) \circ \tilde{\psi} : TU \rightarrow U \times \mathbb{R}^n$$

is a fiber-preserving diffeomorphism.

- The restriction on each fiber

$$\phi|_{T_pM} : T_pM \rightarrow \{p\} \times \mathbb{R}^n$$

is a linear isomorphism.

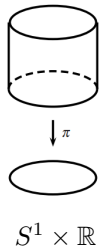
Vector Bundles

Example 2 Given a manifold M , let $\pi : M \times \mathbb{R}^r \rightarrow M$ be the projection to the first factor. Then $M \times \mathbb{R}^r \rightarrow M$ is a vector bundle of rank r , called the **product bundle** of rank r over M .

- The vector space structure on the fiber $\pi^{-1}(p) = \{(p, v) \mid v \in \mathbb{R}^r\}$ is the obvious one:

$$(p, u) + (p, v) = (p, u + v), \quad b \cdot (p, v) = (p, bv), \quad b \in \mathbb{R}.$$

- A local trivialization on $M \times \mathbb{R}^r$ is given by the identity map $1_{M \times \mathbb{R}^r} : M \times \mathbb{R}^r \rightarrow M \times \mathbb{R}^r$.



Vector Bundles

Let $\pi : E \rightarrow M$ be a C^∞ vector bundle. Suppose $(U, \psi) = (U, x^1, \dots, x^n)$ is a chart on M and

$$\phi : E|_U \xrightarrow{\sim} U \times \mathbb{R}^r, \quad \phi(e) = (\pi(e), c^1(e), \dots, c^r(e)),$$

is a trivialization of E over U . Then

$$(\psi \times 1) \circ \phi = (x^1, \dots, x^n, c^1, \dots, c^r) : E|_U \xrightarrow{\sim} U \times \mathbb{R}^r \xrightarrow{\sim} \psi(U) \times \mathbb{R}^r \subset \mathbb{R}^n \times \mathbb{R}^r$$

is a diffeomorphism of $E|_U$ onto its image and so is a chart on E .

We call $x^1, \dots, x^n$ the **base coordinates** and $c^1, \dots, c^r$ the **fiber coordinates** of the chart

$$(E|_U, (\psi \times 1) \circ \phi)$$

on E .

Bundle Map

Let $\pi_E : E \rightarrow M$, $\pi_F : F \rightarrow N$ be two vector bundles, possibly of different ranks.

A **bundle map** from E to F is a pair of maps $(f, \tilde{f})$,

$$f : M \rightarrow N, \quad \tilde{f} : E \rightarrow F,$$

such that

- the diagram

$$\begin{array}{ccc} E & \xrightarrow{\tilde{f}} & F \\ \pi_E \downarrow & & \downarrow \pi_F \\ M & \xrightarrow{f} & N \end{array}$$

is commutative.

- $\tilde{f}$ is linear on each fiber; i.e., for each $p \in M$, $\tilde{f} : E_p \rightarrow F_{f(p)}$ is a linear map of vector spaces.

Bundle Map

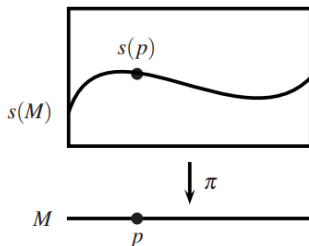
Example A smooth map $f : N \rightarrow M$ of manifolds induces a bundle map $(f, \tilde{f})$, where $\tilde{f} : TN \rightarrow TM$ is given by

$$\tilde{f}(p, v) = (f(p), f_{*,p}(v)) \in \{f(p)\} \times T_{f(p)}M \subset TM$$

for all $v \in T_pN$.

Smooth Sections

A **section** of a vector bundle $\pi : E \rightarrow M$ is a map $s : M \rightarrow E$ such that $\pi \circ s = \mathbf{1}_M$.



We say that a section is smooth if it is smooth as a map from $M \rightarrow E$.

Example) Vector field $X : M \rightarrow TM$, $p \mapsto X_p \in T_pM$.

Smooth Sections

Proposition 12.9

Let s and t be C^∞ sections of a C^∞ vector bundle $\pi : E \rightarrow M$ and let f be a C^∞ real-valued function on M . Then

(i) the sum $s + t : M \rightarrow E$ defined by

$$(s + t)(p) = s(p) + t(p) \in E_p, \quad p \in M,$$

is a C^∞ section of E .

(ii) the product $fs : M \rightarrow E$ defined by

$$(fs)(p) = f(p)s(p) \in E_p, \quad p \in M,$$

is a C^∞ section of E .

Therefore, the set of all C^∞ sections of E is a vector space, $C^\infty(M)$ -module.

Smooth Sections

Proof of (i).

It is clear that $s + t$ is a section of E . To show that it is C^∞ , fix a point $p \in M$ and let V be a trivializing open set for E containing p , with C^∞ trivialization

$$\phi : \pi^{-1}(V) \xrightarrow{\sim} V \times \mathbb{R}^r.$$

Suppose

$$(\phi \circ s)(q) = (q, a^1(q), \dots, a^r(q)), (\phi \circ t)(q) = (q, b^1(q), \dots, b^r(q))$$

for $q \in V$. Because s and t are C^∞ maps, a^i and b^i are C^∞ functions on V . Since ϕ is linear on each fiber,

$$(\phi \circ (s + t))(q) = (q, a^1(q) + b^1(q), \dots, a^r(q) + b^r(q)), \quad q \in V.$$

Therefore $s + t$ is C^∞ at p . □

Smooth Sections

Proof of (ii).

It is clear that fs is a section of E . To show that it is C^∞ , fix a point $p \in M$ and let V be a trivializing open set for E containing p , with C^∞ trivialization $\phi : \pi^{-1}(V) \xrightarrow{\sim} V \times \mathbb{R}^r$.

Suppose

$$(\phi \circ s)(q) = (q, a^1(q), \dots, a^r(q))$$

for $q \in V$. Since s is a C^∞ map, each a^i is a C^∞ function on V . Since ϕ is linear on each fiber,

$$(\phi \circ (fs))(q) = (q, f(q)a^1(q), \dots, f(q)a^r(q)), \quad q \in V.$$

Since f and each a^i are C^∞ , it follows that fs is C^∞ at p . □

Smooth Frames

A **frame** for a vector bundle $\pi : E \rightarrow M$ over an open set U is a collection of sections $s_1, \dots, s_r$ of E over U that at each point $p \in U$, the elements $s_1(p), \dots, s_r(p)$ form a basis for the fiber $E_p = \pi^{-1}(p)$.

A frame $s_1, \dots, s_r$ is said to be smooth C^∞ if $s_1, \dots, s_r$ are C^∞ sections.

Smooth Frames

Example Let M be a manifold and $e_1, \dots, e_r$ the standard basis for $\mathbb{R}^r$. Define $\bar{e}_i : M \rightarrow M \times \mathbb{R}^r$ by $\bar{e}_i(p) = (p, e_i)$. Then $\bar{e}_1, \dots, \bar{e}_r$ is a C^∞ frame for the product bundle $M \times \mathbb{R}^r \rightarrow M$.

Example(The frame of a trivialization) Let $\pi : E \rightarrow M$ be a smooth vector bundle of rank r . If $\phi : E|_U \xrightarrow{\sim} U \times \mathbb{R}^r$ is a trivialization of E over an open set U , then ϕ^{-1} carries the C^∞ frame $\bar{e}_1, \dots, \bar{e}_r$ of the product bundle $U \times \mathbb{R}^r$ to a C^∞ frame $t_1, \dots, t_r$ for E over U :

$$t_i(p) = \phi^{-1}(\bar{e}_i(p)) = \phi^{-1}(p, e_i), \quad p \in U.$$

We call $t_1, \dots, t_r$ the C^∞ frame over U of the trivialization ϕ .

Bump Functions

A **bump function** at q supported in U means any continuous nonnegative function ρ on M that is 1 in a neighborhood of q , with $\text{supp } \rho \subset U$, where

$$\text{supp } f = \text{closure of } \{q \in M \mid f(q) \neq 0\} \text{ in } M$$

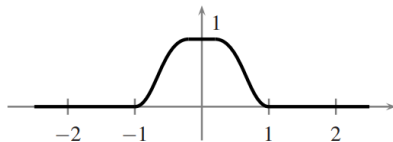


Fig. 13.1. A bump function at 0 on $\mathbb{R}$.

Bump Functions

- C^∞ bump function at 0 in $\mathbb{R}$

Using $f(t) = e^{-1/t}I(t > 0)$, $g(t) = f(t)/\{f(t) + f(1 - t)\}$ with translating, reflecting, scaling, we get

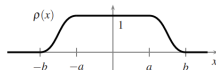


Fig. 13.7. A bump function at 0 on $\mathbb{R}$.

- C^∞ bump function at $\mathbf{0}$ in $\mathbb{R}^n$
Use $\sigma(x) = \rho(\|x\|)$
- C^∞ bump function at q supported in $U \subset M$
For a chart (V, ψ) at p , use $\sigma \circ \psi$

Partition of Unity

A collection $\{A_\alpha\}$ of subsets of a topological space S is said to be **locally finite** if every point $q \in S$ has a neighborhood that meets only finitely many of the sets A_α .

Definition

A C^∞ **partition of unity** on a manifold is a collection of nonnegative C^∞ functions $\{\rho_\alpha : M \rightarrow \mathbb{R}\}_{\alpha \in A}$ such that

- (i) the collection of supports, $\{\text{supp } \rho_\alpha\}_{\alpha \in A}$, is locally finite,
- (ii) $\sum_\alpha \rho_\alpha = 1$.

Given an open cover $\{U_\alpha\}_{\alpha \in A}$ of M , we say that a partition of unity $\{\rho_\alpha\}_{\alpha \in A}$ is *subordinate* to the open cover $\{U_\alpha\}$ if $\text{supp } \rho_\alpha \subset U_\alpha$ for every $\alpha \in A$.

Existence of a Partition of Unity

Proposition 13.6. Let M be a compact manifold and $\{U_\alpha\}_{\alpha \in A}$ an open cover of M . There exists a C^∞ partition of unity $\{\rho_\alpha\}_{\alpha \in A}$ subordinate to $\{U_\alpha\}_{\alpha \in A}$.

Proof

For each $q \in M$, find an open set U_α containing q from the given cover and take ψ_q , a C^∞ bump function at q supported in U_α . Because $\psi_q(q) > 0$, there is a neighborhood W_q of q on which $\psi_q > 0$.

By the compactness of M , the open cover $\{W_q \mid q \in M\}$ has a finite subcover, say $\{W_{q_1}, \dots, W_{q_m}\}$. Let $\psi_{q_1}, \dots, \psi_{q_m}$ be the corresponding bump functions. Then

$$\psi := \sum_i \psi_{q_i}$$

is positive at every point q in M because $q \in W_{q_i}$ for some i .

Existence of a Partition of Unity

Define

$$\phi_i = \frac{\psi_{q_i}}{\psi}, \quad i = 1, \dots, m.$$

Then $\sum \phi_i = 1$ and $\text{supp } \phi_i = \text{supp } \psi_{q_i} \subset U_\alpha$ for some $\alpha \in A$. This shows that $\{\phi_i\}$ is a partition of unity such that for every i , $\text{supp } \phi_i \subset U_\alpha$ for some $\alpha \in A$.

The next step is to make the index set of the partition of unity the same as that of the open cover. For each $i = 1, \dots, m$, choose $\tau(i) \in A$ to be an index such that

$$\text{supp } \phi_i \subset U_{\tau(i)}.$$

We group the collection of functions $\{\phi_i\}$ into subcollections according to $\tau(i)$ and define for each $\alpha \in A$,

$$\rho_\alpha = \sum_{\tau(i)=\alpha} \phi_i.$$

Existence of a Partition of Unity

Then

$$\sum_{\alpha \in A} \rho_{\alpha} = \sum_{\alpha \in A} \sum_{\tau(i)=\alpha} \phi_i = \sum_{i=1}^m \phi_i = 1.$$

Moreover,

$$\text{supp } \rho_{\alpha} \subset \bigcup_{\tau(i)=\alpha} \text{supp } \phi_i \subset U_{\alpha}.$$

So $\{\rho_{\alpha}\}$ is a partition of unity subordinate to $\{U_{\alpha}\}$. □

Existence of a Partition of Unity

Theorem 13.7 (Existence of a C^∞ partition of unity).

Let $\{U_\alpha\}_{\alpha \in A}$ be an open cover of a manifold M .

- (i) There is a C^∞ partition of unity $\{\phi_k\}_{k=1}^\infty$ with every ϕ_k having compact support such that for each k , $\text{supp } \phi_k \subset U_\alpha$ for some $\alpha \in A$.
- (ii) If we do not require compact support, then there is a C^∞ partition of unity $\{\rho_\alpha\}$ subordinate to $\{U_\alpha\}$.

Using partition of unity, we can

- decompose a global object on a manifold into a locally **finite** sum of local objects
- patch together local objects into a global object on a manifold